



ON THE DE RHAM COHOMOLOGY OF CYCLIC COVERS OF $\mathbb{P}^1$

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ALGEBRAIC DE RHAM

k : algebraically closed field of positive characteristic p

X : a smooth projective curve over k

Algebraic de Rham complex:

$$0 \rightarrow \mathcal{O}_X \xrightarrow{d} \Omega_{X/k}^1 \xrightarrow{d} \Omega_{X/k}^2 \xrightarrow{d} \cdots$$

Beware! The "correct" definition for the de Rham cohomology is given through hypercohomology and not the cohomology of the de Rham complex.

Algebraic de Rham cohomology of X is the i -th hypercohomology group of the de Rham complex:

$$H_{\text{dR}}^i(X/k) := \mathbb{H}^i(X, \Omega_{X/k}^\bullet).$$

HOW TO FIND THE k -BASIS

Following the ideas of Bernhard Köck and Joseph Tait from [2] we work as follows:

- Find a k -basis for $H^0(X, \Omega_X)$ (here we use Boseck's results from [1]).
- Use Čech cohomology to write $H^1(X, \mathcal{O}_X)$ as a quotient

$$\frac{\mathcal{O}_X(U_0 \cap U_\infty)}{\{f_0 - f_\infty \mid f_0 \in \mathcal{O}_X(U_0), f_\infty \in \mathcal{O}_X(U_\infty)\}}$$

and find a k -basis.

- Use Čech cohomology to write $H_{\text{dR}}^1(X/k)$ as a quotient.

NOTATION

Let ω be an element of

$$\Omega_\infty := \Omega_X(U_\infty) = \Omega_X(X \setminus \{P_{\infty,1}, \dots, P_{\infty,n}\}).$$

Elements of Ω_∞ are differentials which may have poles only above ∞ . If we see Ω_∞ as a $k(X) := k(x)[y]$ -vector space then its dimension is one.

However as a k -vector space its dimension is infinite, so let $\{\omega_i\}_{i \in I}$ be a k -basis.

We'll use the valuation of any point over $\infty \in \mathbb{P}_k^1$ (say $P_{\infty,1}$, without loss of generality) to "split" ω in two terms:

Let

$$\omega = \sum_{\ell=1}^L \tilde{c}_\ell \omega_{i_\ell}, \tilde{c}_\ell \in k, i_\ell \in I.$$

If there are ω_{i_ℓ} 's with the same valuation at $P_{\infty,1}$, and only then, we replace them with a new term of their sum. We repeat that until we are left with a sum of terms that have pairwise different valuations at $P_{\infty,1}$.

Let the result of this procedure be $\omega = \sum_{\kappa=1}^K c_\kappa \omega_\kappa$, $c_\kappa \in k$.

Define $\omega^{\leq N}$, respectively $\omega^{>N}$, to be the sum of $c_\kappa \omega_\kappa$ such that $-v_{P_{\infty,1}}(\omega_\kappa) \leq N$, respectively $-v_{P_{\infty,1}}(\omega_\kappa) > N$, thus

$$\omega = \omega^{\leq N} + \omega^{>N}.$$

HODGE COHOMOLOGY

Hodge cohomology of X :

$$H_{\text{Hdg}}^\ell(X/k) := \bigoplus_{i+j=\ell} H^i(X, \Omega_{X/k}^j)$$

HODGE-DE RHAM SPLITTING

Spectral sequence connecting the Hodge and de Rham cohomology theories:

$$E_1^{ab} = H^b(X, \Omega_{X/k}^a) \Rightarrow \mathbb{H}^{a+b}(X, \Omega_{X/k}^\bullet)$$

By spectral sequences' theory we get that

$$H_{\text{dR}}^\ell(X/k) \cong H_{\text{Hdg}}^\ell(X/k) \Leftrightarrow$$

the spectral sequence degenerates on E_1 .

For our setup, by a celebrated result from Deligne and Illusie [3], this spectral sequence degenerates on E_1 and gives the following splitting (since we have k -vector spaces):

$$0 \rightarrow H^0(X, \Omega_X) \rightarrow H_{\text{dR}}^1(X/k) \rightarrow H^1(X, \mathcal{O}_X) \rightarrow 0.$$

KUMMER CASE

$$X : y^n = f(x) = \prod_{i=1}^r (x - \rho_i)^{l_i}$$

Assume (w.l.o.g.) that $n \mid l = \deg f(x)$, thus the points above ∞ are not ramified. The points above each ρ_i are ramified with ramification index $e_i = \frac{n}{(n, l_i)}$, and there are $g_i = (n, l_i)$ of them.

Define $\lambda_i = \frac{l_i}{(n, l_i)}$. For $\mu \in \{1, \dots, n-1\}$ we write the Euclidean division

$$\mu \lambda_i = m_i^{(\mu)} e_i + \rho_i^{(\mu)}, \text{ with } 0 \leq \rho_i^{(\mu)} \leq e_i - 1.$$

Define $t^{(\mu)} = \frac{1}{n} \sum_{i=1}^r g_i \rho_i^{(\mu)}$. Set $g_\mu(x) = \prod_{i=1}^r (x - \rho_i)^{m_i^{(\mu)}}$. We define $\phi_\mu(x) := \left(\frac{y^\mu}{g_\mu(x)}\right)^n$ which is a polynomial of x . We also define $\omega_1^{(\mu)} := \frac{y^\mu}{g_\mu(x)} \frac{\phi'_\mu(x)}{n \phi_\mu(x)} dx$ and $\omega_2^{(\mu, \nu)} := \frac{\nu y^\mu}{g_\mu(x)} dx$ which are both in Ω_∞ . Then the elements

$$a_{\mu, \nu} := \left[\left(\frac{\omega_1^{\leq \nu}}{x^\nu} - \frac{\omega_2^{\leq \nu+1}}{x^{\nu+1}}, \frac{\omega_2^{> \nu+1}}{x^{\nu+1}} - \frac{\omega_1^{> \nu}}{x^\nu}, \frac{y^\mu}{x^\nu g_\mu(x)} \right) \right]$$

together with the elements

$$\delta_{\mu, \nu} := \left[\left(\frac{x^{\nu-1} g_\mu(x)}{y^\mu} dx, \frac{x^{\nu-1} g_\mu(x)}{y^\mu} dx, 0 \right) \right]$$

where $1 \leq \mu \leq n-1$ s.t. $t^{(\mu)} \geq 2$ and $1 \leq \nu \leq t^{(\mu)} - 1$, form a basis for $H_{\text{dR}}^1(X/k)$ as a k -vector space.

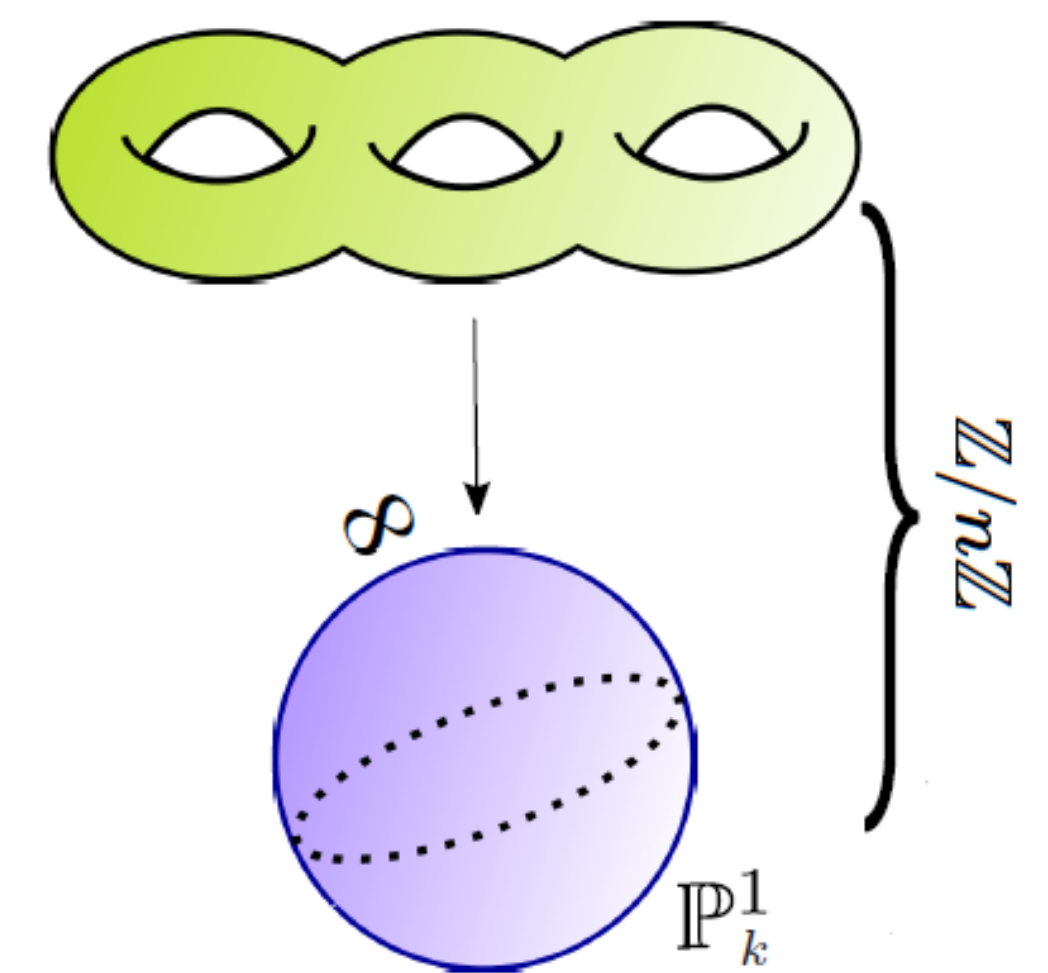
ARTIN-SCHREIER CASE

Everything presented here for the case of Kummer extensions has been carried out also for the case of Artin-Schreier extensions. The calculations and final formulas are somewhat more tedious in this case.

This can be "expected" intuitively as the case where the order of the Galois group isn't prime to the characteristic is always more complicated, and we expect that to be the case when we study the Galois module structure of H_{dR}^1 .

k -BASIS FOR $H_{\text{dR}}^1(X/k)$

When considering curves with function field that is a Galois extension of the rational function field (e.g. with Galois group $\mathbb{Z}/n\mathbb{Z}$, with $(p, n) = 1$, the so-called Kummer extensions) then all the cohomologies that we have described obtain a Galois module structure with respect to the action from the group $\text{Aut}(X)$.



Is it possible to use the splitting of $H_{\text{dR}}^1(X/k)$ to find a k -vector space basis and use it to determine its Galois module structure?

REFERENCES

- [1] Hans Boseck, *Zur Theorie der Weierstraßpunkte*, Mathematische Nachrichten, **19** (1958), 29–63, MR MR0106221 (21 4955).
- [2] Bernhard Köck and Joseph Tait. *On the de-Rham cohomology of hyperelliptic curves*, Res. number theory **4**, 19 (2018). <https://doi.org/10.1007/s40993-018-0111-4>
- [3] P. Deligne and L. Illusie, "Relèvements modulo p^2 et décomposition du complexe de de Rham," *Invent. Math.*, vol. 89, no. 2, pp. 247–270, 1987.

FUTURE WORK

Using the bases that we calculated we intend to find the Galois module structure of H_{dR}^1 under the action of $\text{Aut}(X)$.